Adriel Ventura

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Education

Hofstra University — Hempstead, NY

B.S. Mechanical Engineering, Aerospace Concentration — *Expected Dec. 2027*

- Rabinowitz Honors College · Provost Scholar · CSTEP Scholar
- Coursework: Heat Transfer, Thermodynamics, Fluid Mechanics, Computer-Aided Design, Engineering Materials

Experience

Stellant Systems, Inc. — Jun. 2026 – Present

Mechanical Engineering Co-op — *Melville, NY*

- Support mentor-guided steady-state thermal analysis of RF and defense electronic assemblies in ANSYS Mechanical: geometry clean-up, meshing, boundary-condition debugging, and gradient checks against operating limits
- Revised thermal-pad contact conductance in a thermal model to address artificial temperature gradients
- Designed and fabricated a multi-diameter O-ring cutting fixture through four iterations with technician feedback; applied GD&T, tolerance stack-up, and FDM prototypes to address blade deflection and improve cutting repeatability
- Create and audit Bills of Materials and author Engineering Change Notices (ECNs) for production hardware, routed for cross-functional approval through SharePoint

Projects

CPU Heat Sink Thermal Design | *ENGG 114 Heat Transfer; MATLAB, Python, Fusion 360* — Spring 2026

- Built a MATLAB/Python thermal-resistance model with laminar Nusselt correlations: 0.4485 °C/W and 69.85°C; Fusion 360 FEA predicted 69.775°C
- Chose 15 fins over a cooler 17-fin baseline to widen fin clearance and cut mass 27.5% (75.3 g to 54.6 g); wrote a thermal-test plan (not yet experimentally verified)

NASA L'SPACE Mission Concept Academy | *Thermal Engineer, Team 6 (Artemis)* — Aug. 2026 – Present

- Defined lunar-rover thermal requirements, including a −133 to +121 °C operating envelope (requirement SYS-10)
- Evaluated multilayer insulation, surface coatings, and conductive thermal straps for avionics and payload thermal control; added lunar-dust research (NASA TFAWS 2024) to thermal-risk trade studies

Leadership

Society of Hispanic Professional Engineers (SHPE), Hofstra — May 2026 – Present

President & Chapter Revamp Lead — *Hempstead, NY*

- Rebuilding Hofstra's SHPE chapter as a new student organization: recruited an executive board, drafted the constitution, filed for university recognition through GetInvolvedHU, and held the first Meet the Board event (Oct. 7, 2026)

American Society of Mechanical Engineers (ASME), Hofstra — Apr. 2026 – Present

Co-President & Social Media Manager — *Hempstead, NY*

- Co-lead the executive board with Dara Khaimov; ran steam-turbine and catapult builds and co-hosted the Night Shift Engineering Mixer with SWE and the Commuter Student Association (30 attendees)

Honors & Certifications

1 of 27 participants advanced to the live SHPE x Rockwell Automation AI Hackathon (2026)
Winner, SHPE's Got Talent, Hofstra SHPE (2026)
Nominee, Aviation Week 20 Twenties, Class of 2027 (university nomination; winners announced Jan. 2027)

Technical Skills

Analysis: ANSYS Mechanical (steady-state thermal, meshing), MATLAB, Python, Excel macros, thermal-resistance modeling
CAD: Autodesk Inventor, Siemens NX, SOLIDWORKS, Fusion 360, AutoCAD
Manufacturing: GD&T (ASME Y14.5), tolerance stack-up, DFMA, fixture design, FDM/SLA 3D printing
Documentation: Engineering drawings, BOM creation and audit, ECNs, SharePoint
Data & AI: Databricks, AI-assisted development (Claude), Git